

Problem 1.1: Probability density and current



The time-dependent Schrödinger equation in 3D is

$$i\hbar \frac{\partial}{\partial t} \psi(\mathbf{x}, t) = -\frac{\hbar^2}{2m} \nabla^2 \psi(\mathbf{x}, t) + V(\mathbf{x}) \psi(\mathbf{x}, t).$$

Using the Schrödinger equation, we can calculate the probability density $\rho(\mathbf{x}, t) = |\psi(\mathbf{x}, t)|^2$, which gives the probability of finding the particle at an infinitesimally small region around $\mathbf{x}$ at time t . For a closed system, the probability density should satisfy the continuity equation:

$$\frac{\partial \rho(\mathbf{x}, t)}{\partial t} + \nabla \cdot \mathbf{j}(\mathbf{x}, t) = 0.$$

This is a statement of local conservation of probability, where $\mathbf{j}(\mathbf{x}, t)$ is the probability current density (how much probability is flowing through a unit area per unit time at position $\mathbf{x}$ and time t).

- a) Derive the standard expression for the probability current density:

$$\mathbf{j}(\mathbf{x}, t) = \frac{\hbar}{2mi} (\psi^*(\mathbf{x}, t) \nabla \psi(\mathbf{x}, t) - \psi(\mathbf{x}, t) \nabla \psi^*(\mathbf{x}, t)).$$

- b) Rewrite $\mathbf{j}(\mathbf{x}, t)$ in terms of the momentum operator $\hat{\mathbf{p}} = -i\hbar \nabla$.
- c) In the Schrödinger picture and using Dirac notation, find an expression for the current operator $\hat{j}(\mathbf{x})$ in a basis-independent way. Is this operator time-dependent? What about its expectation value? (hint: recall that $\psi(\mathbf{x}) = \langle \mathbf{x} | \psi \rangle$ and $\psi^*(\mathbf{x}) = \langle \psi | \mathbf{x} \rangle$).
- d) Now consider a polar form for the wave function: $\psi(\mathbf{x}, t) = \sqrt{\rho(\mathbf{x}, t)} e^{iS(\mathbf{x}, t)/\hbar}$, where $\rho(\mathbf{x}, t)$ is the probability density and $S(\mathbf{x}, t)$ is the phase of the wave function. Show that the probability current density can be written as

$$\mathbf{j}(\mathbf{x}, t) = \frac{\rho(\mathbf{x}, t)}{m} \nabla S(\mathbf{x}, t).$$

What does this mean for a *real* wavefunction $\psi(\mathbf{x})$?

- e) For a plane-wave solution of the form $\psi(\mathbf{x}, t) = \frac{1}{\sqrt{V}} e^{i(\mathbf{k} \cdot \mathbf{x} - \omega t)}$, where V is some normalization volume (formally, $V \rightarrow \infty$ for a bona-fide plane wave), $\mathbf{k}$ is the wave vector, and ω is the angular frequency, compute the probability density $\rho(\mathbf{x}, t)$. Is it time-dependent? How about the probability current density $\mathbf{j}(\mathbf{x}, t)$?